

Loan practice speedwork

Rules: You will have *exactly 5 minutes* to answer each question. After the time is up, you will work with a partner to compare solutions.

Problem.

Yoshi is considering a loan of \$400,000 to purchase a home. Two level-payment mortgage options with monthly payments are available:

- Loan A (30-year): nominal annual rate 6.5% compounded monthly.
- Loan B (50-year): nominal annual rate 6.875% compounded monthly.

Calculate the difference in total interest paid over the full term of the loans.

- A. 159,700
- B. 345,000
- C. 511,000
- D. 673,000
- E. 960,000

Problem.

A \$10,000 loan is repaid with equal payments at the end of each year for 30 years. The annual effective interest rate is 5.74%.

Find t such that the interest portion of payment t (the t th payment) is 20% of the total payment.

- A. 23
- B. 24
- C. 25
- D. 26
- E. 27

Problem.

Kevin takes out a 10-year loan of L , which he repays by the amortization method at an annual effective interest rate of i . Kevin makes payments of 1000 at the end of each year.

The total amount of interest repaid during the life of the loan is also equal to L .

Calculate the amount of interest repaid during the first year of the loan.

- A. 725
- B. 750
- C. 755
- D. 760
- E. 765

Problem.

A loan is repaid with level annual payments based on an annual effective interest rate of 7%.

The 8th payment consists of 789 of interest and 211 of principal.

Calculate the amount of interest paid in the 18th payment.

A. 415

B. 444

C. 585

D. 556

E. 612

Problem.

A 10-year loan with an effective annual interest rate of 5% is to be repaid with the following payments:

- (i) 100 at the end of the second year;
- (ii) 200 at the end of the fourth year;
- (iii) 300 at the end of the sixth year;
- (iv) 400 at the end of the eighth year;
- (v) 500 at the end of the tenth year.

Calculate the amount of interest included in the second payment.

- A. 100
- B. 103
- C. 105
- D. 109
- E. 114

Problem.

Caleb purchased his first home for \$420,000. He made a 10% down payment and financed the remaining purchase price.

The terms of the loan were as follows:

- 30-year loan
- Payments made monthly
- Interest charged at 9% convertible monthly

In which month is the first payment for which the principal component is greater than half of the payment?

- A. 267
- B. 269
- C. 270
- D. 266
- E. 268

Solutions:

1. C
2. E
3. C
4. C
5. D
6. B